ModelIt!

Student Activity Pack

Why Some People Can't Drink Milk

Gene Expression, Enzymes, and the Lactose Puzzle

Grade 09L1 | NGSS HS-LS3-1 | Systems & Models

STUDENT NAME _____ DATE _____

ModelIt! — Discovery Collective

© 2026 Discovery Collective. All rights reserved.

Table of Contents

ModelIt!

1 Pre-Assessment 3

2 Scoring Rubric 4

3 Component Analysis 5

4 Model Recording Page 6

5 Simulation Observations 7

6 Research & Extend 8

7 Reflection Questions 9

8 STEM Challenge 10

1 Pre-Assessment

ModelIt!

What Do You Know? Answer each question below to the best of your ability. There are no wrong answers — this helps your teacher understand what you already know!

1 What is an enzyme and what does it do?

2 What does it mean when a gene is 'expressed' or 'active'?

3 Why do some people get stomach problems from drinking milk?

4 How do genes determine physical traits like enzyme production?

2 Scoring Rubric

ModelIt!

Your teacher will use this rubric to assess your work. Read through each level to understand what is expected!

Skill	Expert (4)	Proficient (3)	Developing (2)	Beginning (1)
Identifying Components	EXPERT Correctly identifies and sorts ALL components as internal or external with detailed reasoning	PROFICIENT Correctly identifies and sorts MOST components with some reasoning	DEVELOPING Identifies components but has difficulty sorting or explaining	BEGINNING Needs significant help identifying components
Establishing Relationships	EXPERT Accurately describes ALL positive and negative relationships with clear evidence	PROFICIENT Accurately describes MOST relationships with some evidence	DEVELOPING Identifies some relationships but struggles with direction	BEGINNING Needs help understanding how components connect
Running Simulations	EXPERT Makes accurate predictions AND explains surprising results using evidence	PROFICIENT Makes reasonable predictions and records clear observations	DEVELOPING Runs simulations but predictions or observations are incomplete	BEGINNING Needs help running simulations or recording data
Explaining Results	EXPERT Uses multiple pieces of evidence to explain HOW and WHY systems interact	PROFICIENT Uses evidence to explain system interactions	DEVELOPING Gives partial explanations with limited evidence	BEGINNING Needs help connecting evidence to explanations

3 Component Analysis

ModelIt!

What is a Component?

A **component** is a part of a system that plays a specific role. Components can be **EXTERNAL** (come from outside the system) or **INTERNAL** (happen within the system).

Sort These Components

Lactase Gene Activity | Lactase Enzyme Production | Lactose Breakdown | Gut Bacteria Response

EXTERNAL	INTERNAL

Explain your reasoning: Why did you sort each component the way you did?

4 Model Recording Page

ModelIt!

Draw your model showing all components and their relationships. Use the relationship key below to label your arrows!

+ **POSITIVE:** When A increases, B also increases

− **NEGATIVE:** When A increases, B decreases

Draw your model here

? Explain ONE relationship in your model. What happens when one component changes?

5 Simulation Observations

ModelIt!

Run each scenario in ModelIt! First, write your **prediction**. Then run the simulation and record what you **observe**.

1 Scenario 1: Lactose Tolerant

Lactase Gene Activity: HIGH

MY PREDICTION

WHAT I OBSERVED

2 Scenario 2: Lactose Intolerant

Lactase Gene Activity: LOW

MY PREDICTION

WHAT I OBSERVED

3 Scenario 3: Partial Tolerance

Lactase Gene Activity: MEDIUM

MY PREDICTION

WHAT I OBSERVED

6 Research & Extend

ModelIt!

Real systems are more complex than our model! Here are additional components scientists study. Choose one and explain how it could connect to our model.

Possible New Components

Dairy Amount — The volume of lactose-containing food consumed. Small amounts may be tolerated even with low lactase, but large amounts overwhelm the system.

Gut Microbiome Composition — The specific bacteria species in your large intestine determine how aggressively undigested lactose is fermented and how severe symptoms are.

Epigenetic Regulation — Chemical modifications to DNA (methylation) that silence the lactase gene after weaning. This is the mechanism that 'turns off' lactase in most adults.

My Extension

A Which new component would you add to the model?

B How would it connect to the existing components? (Draw or describe)

C What evidence or reasoning supports your connection?

7 Reflection Questions

ModelIt!

The Big Question

Why can some people drink a milkshake with no problem while others suffer for hours? Use evidence from your simulation to support your answers.

1 Why is lactose TOLERANCE the mutation, not lactose intolerance?

2 How does this example demonstrate the connection between DNA, proteins, and physical traits?

3 Why did the lactase persistence mutation spread in some populations but not others?

4 How does the threshold effect in your model relate to real-world dairy tolerance?

5 What does this tell us about human evolution being driven by cultural practices (dairy farming)?

8 STEM Challenge: The Dairy Product Design Challenge

ModelIt!

YOUR ENGINEERING MISSION

Design a dairy product line that people with lactose intolerance can enjoy without symptoms, using your understanding of enzyme science.

THE SCENARIO

A food company wants to create a line of dairy products for the 65% of the world's population that is lactose intolerant. Using your model of lactase-lactose interactions, design products that solve the digestion problem through different scientific approaches.

Engineering Design Thinking

1. WHAT SCIENTIFIC APPROACH WILL YOUR PRODUCT USE TO SOLVE THE LACTOSE PROBLEM?
2. HOW WILL YOU ENSURE THE PRODUCT STILL TASTES AND FEELS LIKE DAIRY?
3. WHAT TESTING WOULD YOU NEED TO VERIFY YOUR PRODUCT WORKS FOR INTOLERANT CONSUMERS?
4. HOW DOES YOUR SOLUTION COMPARE TO EXISTING PRODUCTS LIKE LACTAID?

My Design

DESIGN AREA — DRAW YOUR SOLUTION

N

! Explain your design using scientific evidence from the simulation:

Real Career Connection

Genetic Counselors help people understand how their genes affect their health, from lactose intolerance to disease risk. Food Scientists apply enzyme chemistry to create products like lactose-free milk, yogurt, and cheese — a market worth over $15 billion globally.